(iii) Transition metal hydrides are used as powerful reducing agents

(iv) Metallic carbides are chemically inert.

4.3.10 Formation of complexes

Transition elements have a tendency to form coordination compounds with a species that has an ability to donate an electron pair to form a coordinate covalent bond. Transition metal ions are small and highly charged and they have vacant low energy orbitals to accept an electron pair donated by other groups. Due to these properties, transition metals form large number of complexes. Examples: $[Fe(CN)_6]^{4-}$, $[Co(NH_3)_6]^{3+}$, etc..

The chemistry of coordination compound is discussed in unit 5.

4.4 Important compound of Transition elements

Oxides and Oxoanions of Metals

Generally, transition metal oxides are formed by the reaction of transition metals with molecular oxygen at high temperatures. Except the first member of 3d series, Scandium, all other transition elements form ionic metal oxides. The oxidation number of metal in metal oxides ranges from +2 to +7. As the oxidation number of a metal increases, ionic character decreases, for example, Mn_2O_7 is covalent. Mostly higher oxides are acidic in nature, Mn_2O_7 dissolves in water to give permanganic acid ($HMnO_4$) , similarly CrO_3 gives chromic acid (H_2CrO_4) and dichromic acid ($H_2Cr_2O_7$). Generally lower oxides may be amphoteric or basic, for example, Chromium (III) oxide - Cr_2O_3, is amphoteric and Chromium(II) oxide, CrO, is basic in nature.

Potassium dichromate $K_2Cr_2O_7$

Preparation:

Potassium dichromate is prepared from chromite ore. The ore is concentrated by gravity separation. It is then mixed with excess sodium carbonate and lime and roasted in a reverbratory furnace.

$$4\,FeCr_2O_4 + 8\,Na_2CO_3 + 7\,O_2 \xrightarrow{900 - 1000^0C} 8\,Na_2CrO_4 + 2\,Fe_2O_3 + 8\,CO_2\uparrow$$

The roasted mass is treated with water to separate soluble sodium chromate from insoluble iron oxide. The yellow solution of sodium chromate is treated with concentrated sulphuric acid which converts sodium chromate into sodium dichromate.

$$\underset{\substack{\text{sodium chromate}\\\text{(yellow)}}}{2\,Na_2CrO_4} + H_2SO_4 \longrightarrow \underset{\substack{\text{sodium dichromate}\\\text{(orange red)}}}{Na_2Cr_2O_7} + Na_2SO_4 + H_2O$$

The above solution is concentrated to remove less soluble sodium sulphate. The resulting solution is filtered and further concentrated. It is cooled to get the crystals of $Na_2SO_4.2H_2O$.

The saturated solution of sodium dichromate in water is mixed with KCl and then concentrated to get crystals of NaCl. It is filtered while hot and the filtrate is cooled to obtain $K_2Cr_2O_7$ crystals.

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